MILITARY SPECIFICATION

NUT, CLINCH

This specification has been approved by the Naval Ordnance Systems Command, Department of the Navy.

1. SCOPE AND CLASSIFICATION

1.1 Scope - This specification covers the general requirements for clinch corrosion resisting steel nuts used to provide a wear resistant thread in soft metals.

1.2 Classification - Clinch nut shall be of the following type and style (see 6.2).

Type I Flush
Style 1 Hex

1.2.1 Sizes - The nuts shall be classified according to thickness and by the nominal internal thread size and type as specified in the National Bureau of Standards (NBS) Handbook H-28, Part I.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of the specification to the extent specified herein.

SPECIFICATIONS

Federal

QQ-P-35	Passivation Treatments for Austenitic, Ferritic, and Martensitic Corrosion-Resisting Steel (Fastening Devices)

Military

MIL-H-3982	Hardware (Fasteners and Related Items); Packaging and Packing for Shipment and Storage of

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STANDARDS

Federal

FED-STD-66	Steel; Chemical Composition and Hardenability

Military

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes
MS 51957	Screw, Machine-Pan Head, Cross-Recessed, Corrosion Resisting Steel, Passivated, UNC-2A
MS 51958	Screw, Machine-Pan Head, Cross, Resessed, Corrosion Resisting Steel, Passivated, UNF-2A
MS3214(OS)	Nut, Plain, Clinch-Flush

2.2 Other publications The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated the issue in effect on date of invitation for bids or request for proposal shall apply

USAS1 B46.1	Surface Texture; Surface Roughness, Waviness, and Lay.

National Bureau of Standards Handbook H-28, Part 1 - Screw-Thread Standards for Federal Services.

(Copies of specifications, standards and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.

3. REQUIREMENTS

3.1 The end product Unless otherwise specified (see 6.2), the clinch nut furnished under this specification shall be a product which has been inspected and passed the conformance inspection specified herein.

3.2 Design and construction The clinch nut as shown on MS3214(OS) shall have a hexagon head to resist pullout and torqueout when installed flush in the sheet parent material. In addition the nut shall have a recessed neck cavity designed to accomodate the displaced parent material, when installed, to resist pushout.

3.3 Material The nuts shall be manufactured of AISI 303 corrosion resisting steel as specified in Fed. Std. No. 66.

3.4 Dimensions and tolerances Dimensions and tolerances after finishing shall be as specified on the applicable standard. The nuts shall also meet the dimensional requirements of 3.4.1

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3.4.1 Concentricity - Opposing hex flats shall be equally concentric with the tapped hole within 0.010 inch total indicator reading.

3.5 Threads - Threads shall be class 2B UNC or UNF thread series, as specified (see 6.2). The dimensions and tolerances of thread shall conform to National Bureau of Standards (NBS) Handbook H-28, Part I.

3.6 Surface - The surface roughness of all external surfaces shall not exceed 125 microinches roughness height rating (RHR). RHR value shall be interpreted in accordance with USAS B-46.1.

3.7 Protective finish - The nuts shall be passivated, following fabrication, in accordance with QQ-P-35.

3.8 Performance - The clinch nuts shall conform to the values of push out force, pull out force, and torque out moment delineated for each size in Table II, when tested in accordance with the methods described in paragraphs 4.6.2, 4.6.3 and 4.6.4.

3.9 Workmanship - The nuts shall be uniform in quality and shall be free from irregularities, burrs, or slivers which could adversely affect safety or performance.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Classification of inspection - The inspection of clinch nuts shall be classified as follows:

(a) Quality conformance inspection

4.3 Quality conformance inspection - Quality conformance inspection shall consist of the following inspections:

Inspection	Samples tested
(a) Visual and dimensional	All
(b) Push out	3
(c) Pullout	3
(d) Torque out	3

4.4 Test Methods -

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4.4.1 Quality conformance testing Quality conformance testing shall consist of the following tests conducted in the order listed and utilizing the same samples (see 4.5.2) for conducting all tests:

(a) Visual and dimensional (b) Push out (4.6.2)
(c) Pull out (4.6.3) (d) Torque out (4.6.4)

4.5 Sampling

4.5.1 Inspection lot A lot shall consist of finished clinch nuts of the same material, type, style, and size produced under essentially like conditions. Presentation of lots shall be in accordance with MIL-STD-105.

4.5.2 Sampling for quality conformance inspection Unless otherwise specified, the contractor shall furnish production samples for inspection to determine conformance with this specification. The production samples shall consist of nine (9) of each item on contract or order. The samples shall be identified with the manufacturers part number and contract number and shall be inspected as specified herein and in accordance with the contract or order (see 6.2).

4.5.3 Sampling for acceptable quality level A random sample of clinch nuts shall be taken from each lot in accordance with MIL-STD-105, inspection level S-1. The acceptable quality level shall be 2.5 percent.

4.5.4 Sampling for packaging and packing Sampling for inspection of packaging and packing shall be in accordance with MIL-H-3982.

4.6 Examination

4.6.1 Visual and dimensional Each clinch nut taken at random as specified in 4.3, 4.5.2, or 4.5.3, as applicable, shall be examined to verify conformance to requirements of this specification. Examination shall be conducted in accordance with Table I. Any nut having one or more defects shall be rejected.

4.6.2 Push out test The sample of nuts selected at random as specified in 4.3 and 4.5.2 shall be inserted in mounting holes in 5052 H-34 aluminum sheet of the same nominal thickness as the test nuts in accordance with Table II for the nuts being tested. The nuts shall then be installed by the application of a steadily increasing pressure by means of an arbor press, vise, etc., until the top edge of the nut hex is flush with the top of the aluminum sheet. Stainless steel screws, in accordance with MS 51957 or MS 51958, as applicable, shall be threaded through the nut from the hex surface. A minimum of two (2) complete threads shall extend beyond the bottom surface of the nut. A force shall then be applied under the screw head as indicated in Figure I, and the force necessary to dislodge the nut from the sheet shall be measured. The force required must not be less than the values shown on Table II for the sizes tested.

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TABLE I

CLASSIFICATION OF DEFECTS, END ITEM

CATEGORIES	DEFECTS	METHOD
Critical	None defined	
Major		
101	Nut sizes	SIE[1]
102	Nut height	SIE
103	Hex dimension	SIE
104	Nut bottom diameter	SIE
105	Thread series and class	SIE
106	Surface finish	SIE
107	Passivation omitted	Visual
Minor		
201	Concentricity of Hex Flats	SIE

1 Standard inspection equipment

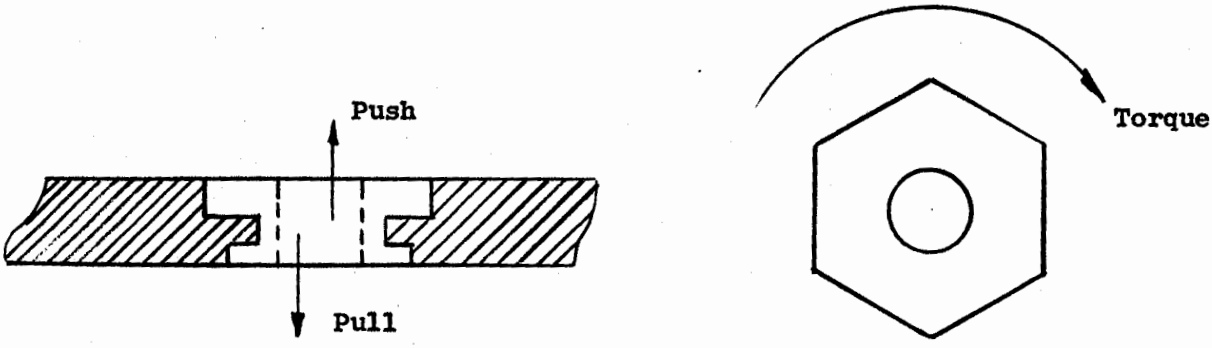


FIGURE I

Application of force for tests

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TABLE II

MINIMUM ACCEPTABLE TEST VALUES BY SIZE

THICKNESS	SIZE & THREAD	HEX	*MOUNTING HOLE DIAM ±.002	+PUSH OUT FORCE lbs.	+PULL OUT FORCE lbs.	+TORQUE OUT in-lbs.
.058	2-56	3/16	.161	175	400	15
		1/4	.203	200	500	25
	4-40	3/16	.161	175	400	15
		1/4	.203	200	500	24
.085	4-40	3/16	.161	250	700	24
		1/4	.203	350	800	50
	6-32	1/4	.203	350	800	50
		5/16	.265	400	1000	65
	8-32	5/16	.265	400	1000	65
	10-32	5/16	.265	400	1000	65
	1/4-20	7/16	.386	450	1250	200
	1/4-28	7/16	.386	450	1250	200
.117	4-40	3/16	.161	250	725	24
		1/4	.203	350	830	50
	6-32	1/4	.203	350	830	50
		5/16	.265	400	1050	65
	8-32	5/16	.265	400	1050	65
	10-32	5/16	.265	400	1050	65
	1/4-20	7/16	.386	450	1250	200
	1/4-28	7/16	.386	450	1250	200
.147	6-32	1/4	.203	350	850	50
		5/16	.265	400	1100	65
	8-32	5/16	.265	400	1100	65
	10-32	5/16	.265	400	1100	65
	1/4-20	7/16	.386	450	1250	200
	1/4-28	7/16	.386	450	1250	200
.177	6-32	1/4	.203	350	850	50
		5/16	.265	400	1150	65
	8-32	5/16	.265	400	1150	65
	10-32	5/16	.265	400	1150	65
	1/4-20	7/16	.386	450	1250	200
	1/4-28	7/16	.386	450	1250	200
.236	6-32	1/4	.203	350	850	50
		5/16	.265	400	1150	65
	8-32	5/16	.265	400	1150	65
	10-32	5/16	.265	400	1150	65
	1/4-20	7/16	.386	450	1250	200
	1/4-28	7/16	.386	450	1250	200

* Hole may be punched or drilled. Do not ream or deburr edges.

+ Figures based on installation in 5052 H-34 aluminum sheet.

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4.6.3 Pull out test The sample of nuts selected at random as specified in 4.3 and 4.5.2 shall be installed in 5052 H-34 aluminum sheet as described in 4.6.2. Stainless steel screws in accordance with MS51957 or MS51958, as applicable, shall be threaded through the nut from the bottom. A minimum of two (2) complete threads shall extent beyond the hex surface of the nut. A force shall be applied under the screw head as indicated in Figure I, and the force necessary to dislodge the nut from the sheet shall be measured. The force required must not be less than values shown in Table II for the sizes tested.

4.6.4 Torque out test The sample of nuts selected at random as specified in 4.3, 4.5.2 and 4.5.3 shall be installed in 5052 H-34 aluminum sheet as described in 4.6.2. Stainless steel screws, in accordance with MS51957 or MS51958, as applicable, shall be threaded through the nut from the hex side and seated hand tight against the nut. Using a torque driver, the screw shall be tightened until either the thread strips or the nut rotates in the sheet parent metal and the torque measured. The nut shall not rotate at torque values below those shown on Table II for the sizes tested. Stripping of the nut thread shall not be considered as a failure of the nut for this test since nut thickness is governed by the required panel thickness and not optimum nut design.

4.6.5 Packaging and packing Examination and test of preservation, packaging and packing shall be in accordance with MIL-H-3982.

5. PREPARATION FOR DELIVERY

5.1 Packaging, packing and marking Packaging, packing and marking shall conform to the requirements of MIL-H-3982 (see 6.2).

6. NOTES

6.1 Intended use - The clinch nuts covered by this specification are intended to provide strong wear-resistant threads in aluminum or brass sheet metal chassis and panels for mounting components in integral torpedo tracking systems.

6.2 Information for contracting officer Contracts or orders should specify the following:

(a) Title number and date of this specification
(b) Type, style, size, thread and length (see 1.2 and 3.5) - and quantity
(c) Applicable standard (see 3.2)
(d) Level of packaging and packing (see 5.1)

Preparing Activity
Navy - (OS)

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